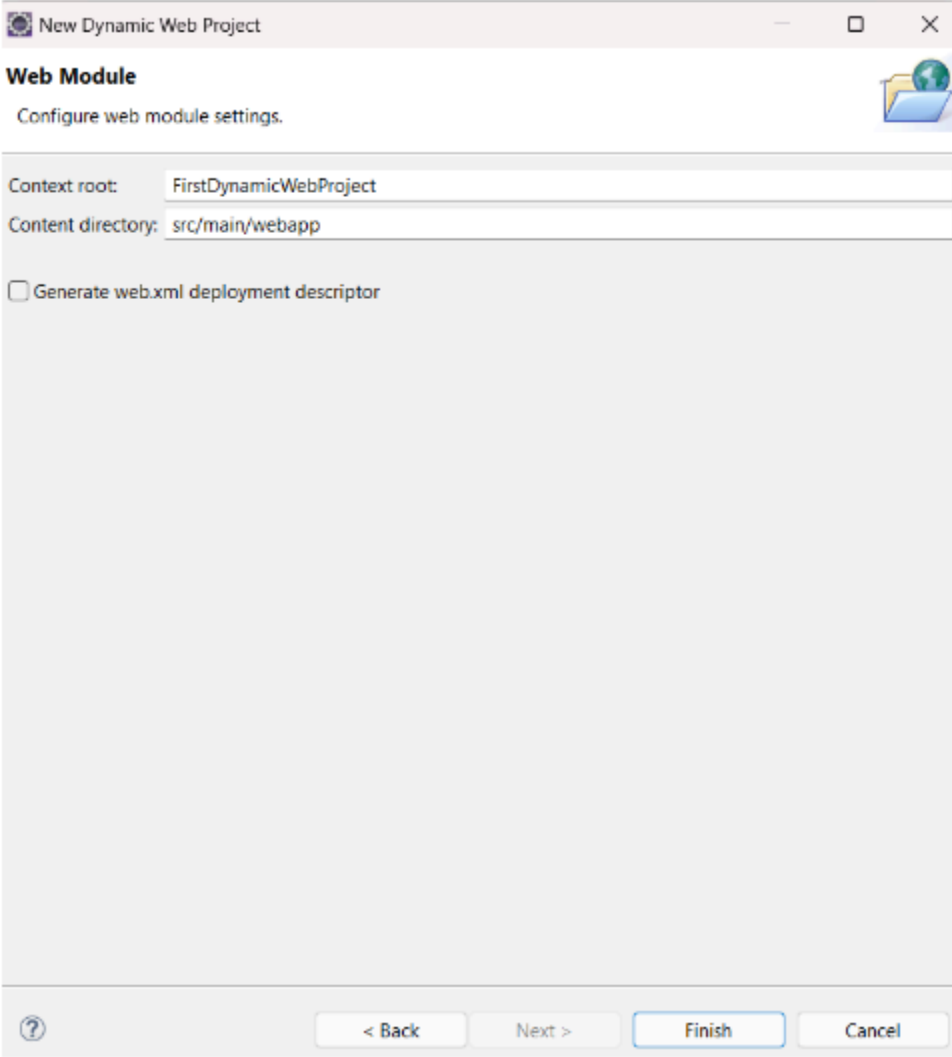


First Servlet WebApp with Dynamic Response

Note: If you do not stop the server and try to run another application, it will show an error that the port is busy.

👉 **Create First WebApp with Servlet**

Step 1: Create a web app project without *web.xml*.



New Dynamic Web Project

Web Module
Configure web module settings.

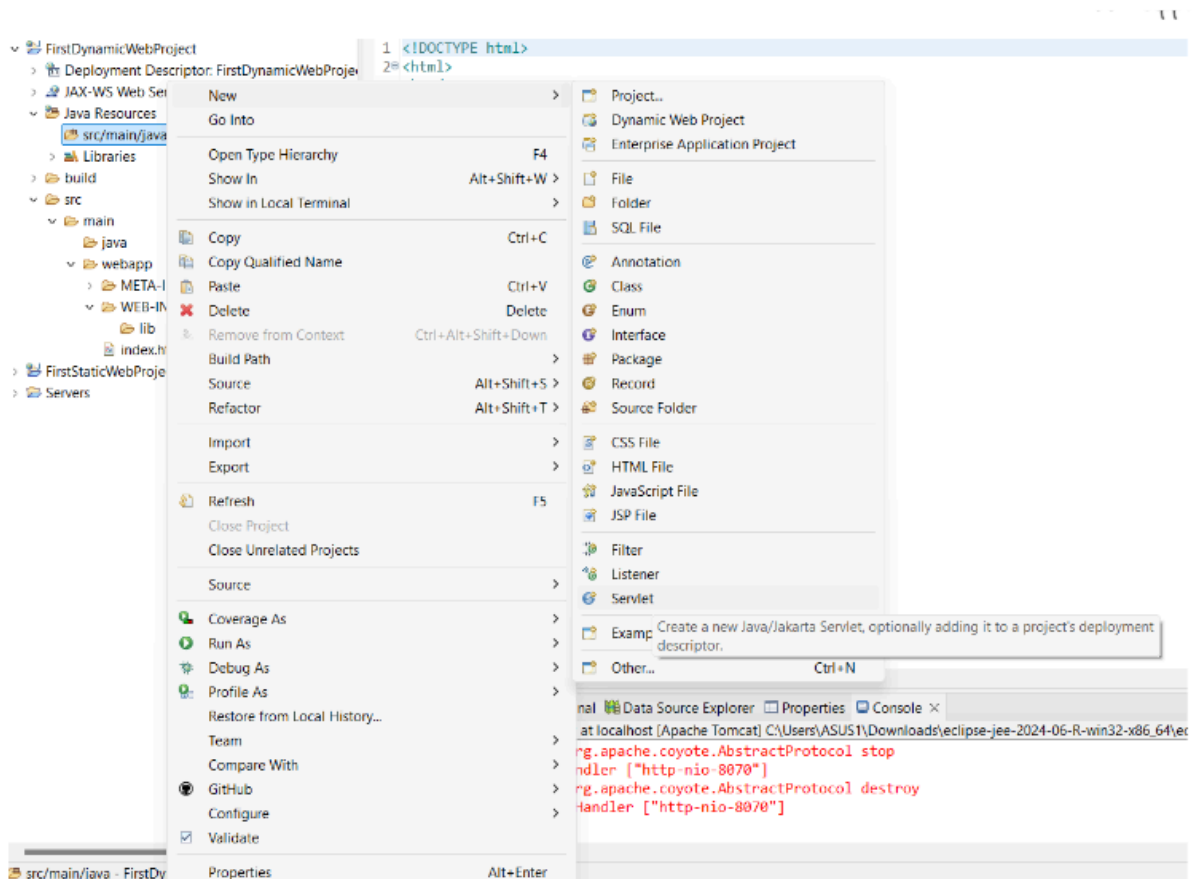
Context root: FirstDynamicWebProject

Content directory: src/main/webapp

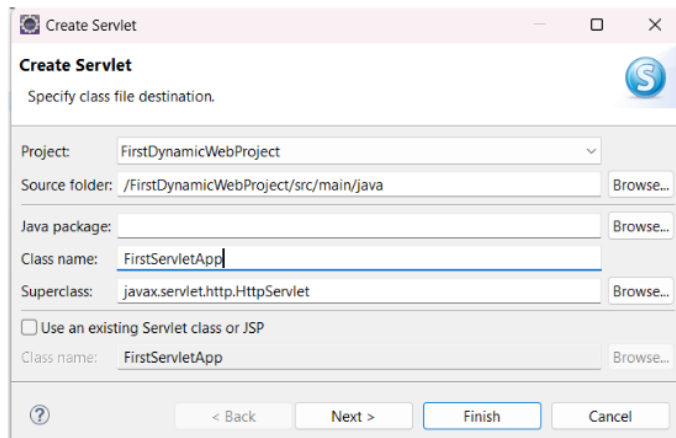
☐ Generate web.xml deployment descriptor

? < Back Next > Finish Cancel

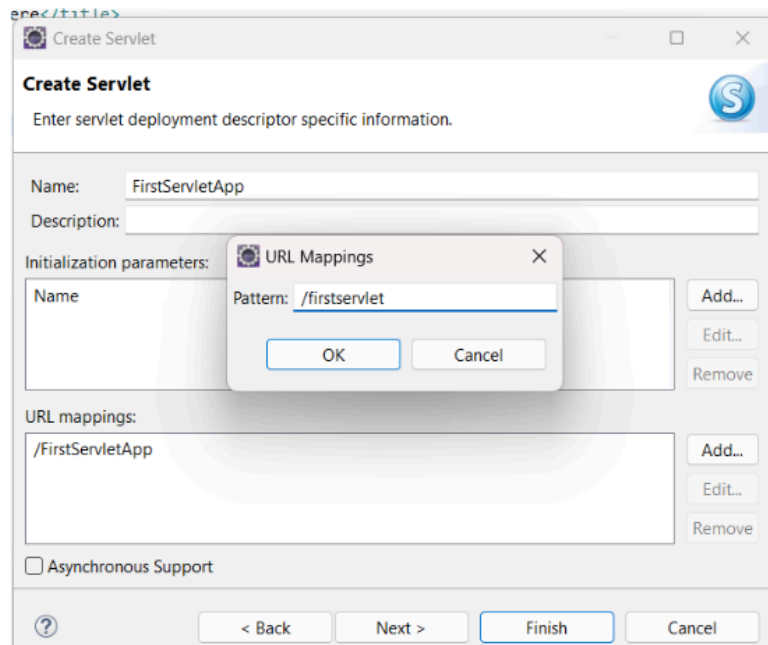
Step 2: In src/main/java, create a servlet inside a package or in the default package.



- Add class name



- Add routing path



- For the first servlet project, select only this option.

Create Servlet

Specify modifiers, interfaces to implement, and method stubs to generate.

Modifiers: ☒ public ☐ abstract ☐ final

Interfaces:

Which method stubs would you like to create?

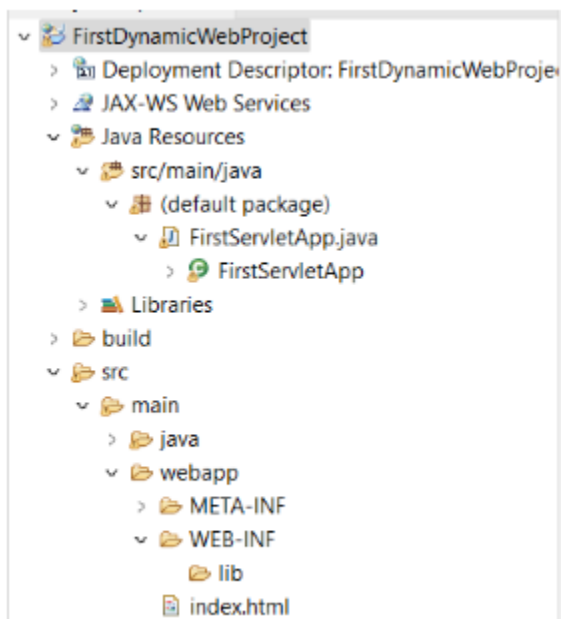
☒ Constructors from superclass
☒ Inherited abstract methods

☐ init ☐ destroy ☐ getServletConfig
☐ getServletInfo ☐ service ☐ doGet
☒ doPost ☐ doPut ☐ doDelete
☐ doInit ☐ doOptions ☐ doTrace

< Back Next > Finish Cancel

Make finish.

👉 Project Structure:



Code:

Index.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body style="font-family: Arial, sans-serif; background-color: #f4f4f9;
display: flex; justify-content: center; align-items: center; height:
100vh; margin: 0;">
  <form style="background-color: #ffffff; padding: 20px;
border-radius: 8px; box-shadow: 0 0 10px rgba(0, 0, 0, 0.1); width:
300px;" action="/firstservlet" method="post">
    <h2 style="text-align: center; color: #333333;">User
Information</h2>

    <div style="margin-bottom: 15px;">
      <label for="uname" style="display: block; font-weight: bold;
margin-bottom: 5px; color: #555555;">Name:</label>
      <input type="text" id="uname" name="uname" style="width:
100%; padding: 10px; border: 1px solid #cccccc; border-radius: 4px;">
    </div>
    <div style="margin-bottom: 20px;">
      <label for="ucity" style="display: block; font-weight: bold;
margin-bottom: 5px; color: #555555;">City:</label>
      <input type="text" id="ucity" name="ucity" style="width:
100%; padding: 10px; border: 1px solid #cccccc; border-radius: 4px;">
```

```
</div>
  <input type="submit" value="Submit" style="width: 100%;
padding: 10px; background-color: #5cb85c; color: white; border: none;
border-radius: 4px; font-size: 16px; cursor: pointer;">
</form>
</body>
</html>
```

FirstServletApp.java

```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/firstservlet")
public class FirstServletApp extends HttpServlet {

    public FirstServletApp() {
        System.out.println("Internally constructor called");
    }

    public void doPost(HttpServletRequest request,
        HttpServletResponse response) throws ServletException, IOException
    {

        String name=request.getParameter("uname");
```

```
String city=request.getParameter("ucity");

PrintWriter pw=response.getWriter();
pw.println("Name is : "+name);
pw.println("City is : "+city);

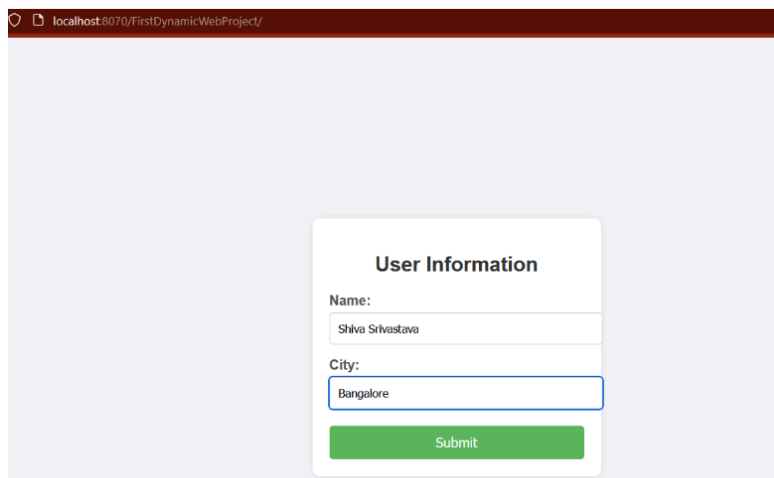
//use html tag for formating
// pw.println("<h1> Name is : "+name+"</h1>");
// pw.println("<h1> City is : "+city+"</h1>");

pw.close();

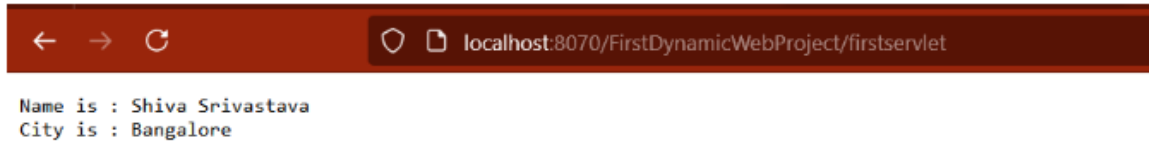
}

}
```

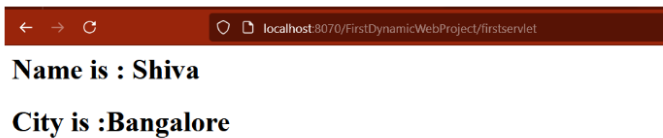
Output:



The screenshot shows a web browser window with the address bar displaying 'localhost:8070/FirstDynamicWebProject/'. The main content area has a light purple background. In the center, there is a white rounded rectangle titled 'User Information'. Inside this rectangle, there are two labels: 'Name:' and 'City:'. Below 'Name:' is a text input field containing 'Shiva Srivastava'. Below 'City:' is a text input field containing 'Bangalore'. At the bottom of the white rectangle is a green button with the text 'Submit'.



Uncomment the HTML tag.



👉 Servlet Concepts and Annotations

@WebServlet Annotation

The @WebServlet annotation is used to declare a servlet.

Example:

```
import jakarta.servlet.annotation.WebServlet;  
import jakarta.servlet.http.HttpServlet;  
@WebServlet("/firstservlet")  
public class FirstServletApp extends HttpServlet {  
  
}
```


Request and Response Objects

- **HttpServletRequest**: Represents the client's request. It contains information such as request parameters, headers, and attributes.
 - Methods:
 - `getParameter(String name)`: Retrieves the value of a request parameter.
 - `getParameter(String city)`: Retrieves the value of a request parameter.

```
String name=request.getParameter("uname");  
String city=request.getParameter("ucity");
```

- **HttpServletResponse**: Represents the server's response. It allows you to set the response status, headers, and content.
 - Methods:
 - `getWriter()`: Returns a `PrintWriter` object to send character text to the client.

```
PrintWriter out = response.getWriter();
```

HttpServlet

- `HttpServlet` is a class provided by Java EE that extends `GenericServlet` and provides methods to handle HTTP requests.
- To create a servlet, extend the `HttpServlet` class.
- Override methods like `doGet`, `doPost`, `doPut`, etc., to handle specific types of HTTP requests.

```
public class FirstServletApp extends HttpServlet {
```

```

@Override
protected void doGet(HttpServletRequest request,
HttpServletRequest response) throws IOException {
    // Handle GET request
}
}

```

Constructor in Servlets

- The constructor of a servlet is called when the servlet is first loaded into memory.

```

public class FirstServletApp extends HttpServlet {
    public FirstServletApp () {
        // Constructor code
    }
}

```

PrintWriter

- PrintWriter is used to send text data to the client.
- It is obtained from the HttpServletResponse object.
- Example:

```

public void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
    String name=request.getParameter("uname");
    String city=request.getParameter("ucity");

    PrintWriter pw=response.getWriter();

```

```
pw.println("Name is : "+name);
```

```
pw.println("City is : "+city);
```

```
//use html tag for formating
```

```
// pw.println("<h1> Name is : "+name+"</h1>");
```

```
// pw.println("<h1> City is : "+city+"</h1>");
```

```
pw.close();
```

```
}
```